Black-Scholes modeling

Black-Scholes modeling relies on a precise framework, one aspect of which is the assumption of a constant, deterministic volatility parameter. The so-called volatility smile observed in historical data was one of the first hints that this assumption is unrealistic. This motivated the introduction of stochastic volatility models. Enriching the model, however, comes with difficulties that are either technical or structural, such as market incompleteness. The goal of this project is to review the main components of Black-Scholes modeling and of stochastic volatility models. Part of the project will be devoted to the most recent aspects of volatility, in particular its roughness. Numerical simulations will be required.

Anthony Réveillac anthony.reveillac@insa-toulouse.fr

Aude Rondepierre Aude.Rondepierre@insa-toulouse.fr

Cathy Maugis-Rabusseau maugis@insa-toulouse.fr